

Actor-Critic and A2C

First, a story

You are learning the piano. Your teacher has two ways of teaching.

Method 1 (REINFORCE): you play the whole piece; the teacher grades at the end — “85 overall”. Can that single number tell you which note was wrong? No. The chord that slipped in bar 3 and the tempo that ran away in bar 12 all drown in the single “85”. You can only guess your way to improvement.

Method 2 (Actor-Critic): the teacher sits beside you, offering a comment every few notes — “too fast here”, “that chord is wrong”, “nicely phrased”. Immediate, localized feedback; progress is much faster.

But a new problem appears: if, after three notes, the teacher decides “I’ve got this student’s level all figured out”, stops listening carefully and just says “fine” from then on — from that moment your practice loses its feedback and progress stops dead.

That is the essence of naive Actor-Critic’s failure — not a wrong direction, but the policy collapsing prematurely (without entropy protection) into a deterministic one, after which the critic fits the value of this very poor policy nearly perfectly, the advantage signal goes to zero, and the actor’s gradient vanishes.

The core problem of this chapter: **how do we keep the critic’s learning rate matched to the actor’s pace of improvement?**

From REINFORCE to Actor-Critic: why replace the return

REINFORCE’s policy-gradient estimator:

$$\nabla_{\theta} J(\theta) = \mathbb{E}_{\tau \sim \pi_{\theta}} \left[\sum_{t=0}^T \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \cdot G_t \right]$$

with $G_t = \sum_{k=t}^T \gamma^{k-t} r_k$ the discounted reward from step t to the end (the Monte Carlo return).

Where the high variance comes from: G_t depends on all future random actions and rewards after t . The longer the episode, the noisier G_t — judging “was the fingering in bar 3 right” by “the final grade of the whole piece” has a terrible signal-to-noise ratio.

The core substitution: let $V^{\pi}(s)$ provide a baseline, replacing G_t by “how much better than average”:

$$G_t \longrightarrow A^{\pi}(s_t, a_t) = Q^{\pi}(s_t, a_t) - V^{\pi}(s_t)$$

This is the **advantage function**. $A > 0$ means “this action is better than average”; $A < 0$ means “worse”. Two benefits: (1) subtracting the action-independent baseline V cuts variance; (2) it measures relative quality, sharpening the signal.

The Advantage Function and the TD Error

Formalization

Let the current policy be π_{θ} and the parameterized value be $V_{\phi}(s)$:

- **State value:** $V^{\pi}(s) = \mathbb{E}_{\pi} \left[\sum_{k=0}^{\infty} \gamma^k r_{t+k} \mid s_t = s \right]$
- **Action value:** $Q^{\pi}(s, a) = \mathbb{E}_{\pi} \left[\sum_{k=0}^{\infty} \gamma^k r_{t+k} \mid s_t = s, a_t = a \right]$
- **Advantage:** $A^{\pi}(s, a) = Q^{\pi}(s, a) - V^{\pi}(s)$

Note $\mathbb{E}_{a \sim \pi} [A^{\pi}(s, a)] = 0$ — the advantage has zero mean by construction: good and bad actions sit symmetrically on either side of zero.

The one-step TD error as an advantage estimate

Computing the true $Q^\pi(s, a)$ needs the full future reward sequence. A feasible approximation:

$$Q^\pi(s_t, a_t) \approx r_t + \gamma V^\pi(s_{t+1}) \quad (\text{one-step bootstrap})$$

Substituting into the advantage definition gives the **one-step TD error**:

$$\delta_t = r_t + \gamma V_\phi(s_{t+1}) - V_\phi(s_t)$$

In plain words: δ_t is “the reward actually received plus the new expectation of the future” minus “the old expectation at departure”. More $\Rightarrow$ this action beat expectations (positive advantage); less $\Rightarrow$ below (negative).

Where the bias comes from: δ_t replaces the true V^π with the approximate V_ϕ , adding function-approximation error (bias). That is the essential trade-off against Monte Carlo: **lower variance, but no longer unbiased**.

Why naive Actor-Critic fails

Four fault modes

The most direct implementation: replace REINFORCE’s per-episode update with per-step online updates and G_t with δ_t . The idea is right, but four latent problems lurk:

1. **Fault 1 — one-step TD advantage, variance still high**

δ_t looks only one step ahead, buffeted by both random rewards and V_ϕ ’s estimation error. Before V_ϕ converges, δ_t is extremely unstable and the actor’s gradient is noisy.

2. **Fault 2 — the two-timescale condition is violated**

Konda & Tsitsiklis (2003) proved: Actor-Critic convergence requires the critic’s learning rate α_w to be much larger than the actor’s α_θ , i.e. $\alpha_\theta/\alpha_w \rightarrow 0$. Naive implementations use one learning rate for both, so the actor updates before the critic has converged, computing gradient directions from a wrong V_ϕ .

3. **Fault 3 — no entropy regularization, premature collapse**

Without an entropy bonus the actor piles probability onto some suboptimal action and becomes near-deterministic. Once the action distribution has no randomness, the policy gradient cannot drive exploration even when δ_t is occasionally nonzero.

4. **Fault 4 — no gradient clipping, parameters can explode**

Single-step $|\delta_t|$ is occasionally huge (badly initialized V_ϕ), causing enormous updates that push the network into unrecoverable regions.

The collapse mechanism: advantage vanishing $\Rightarrow$ actor freezes

Once Fault 3 triggers, the four faults couple into a self-reinforcing loop:

no entropy $\Rightarrow \pi$ becomes deterministic rapidly $\Rightarrow$ every episode repeats the same short trajectory

$\Rightarrow V_\phi$ fits those trajectories quickly ($\delta_t \approx 0$) $\Rightarrow \nabla_\theta J \approx 0 \Rightarrow$ the actor freezes forever.

This is a **stable but catastrophic fixed point**: the more “perfect” the critic, the less incentive the actor has to move; the more the actor freezes, the more solid the critic. The system locks itself in — almost identically across seeds.

Fig 1. Naive Actor-Critic: Unstable Learning (All 5 Seeds)

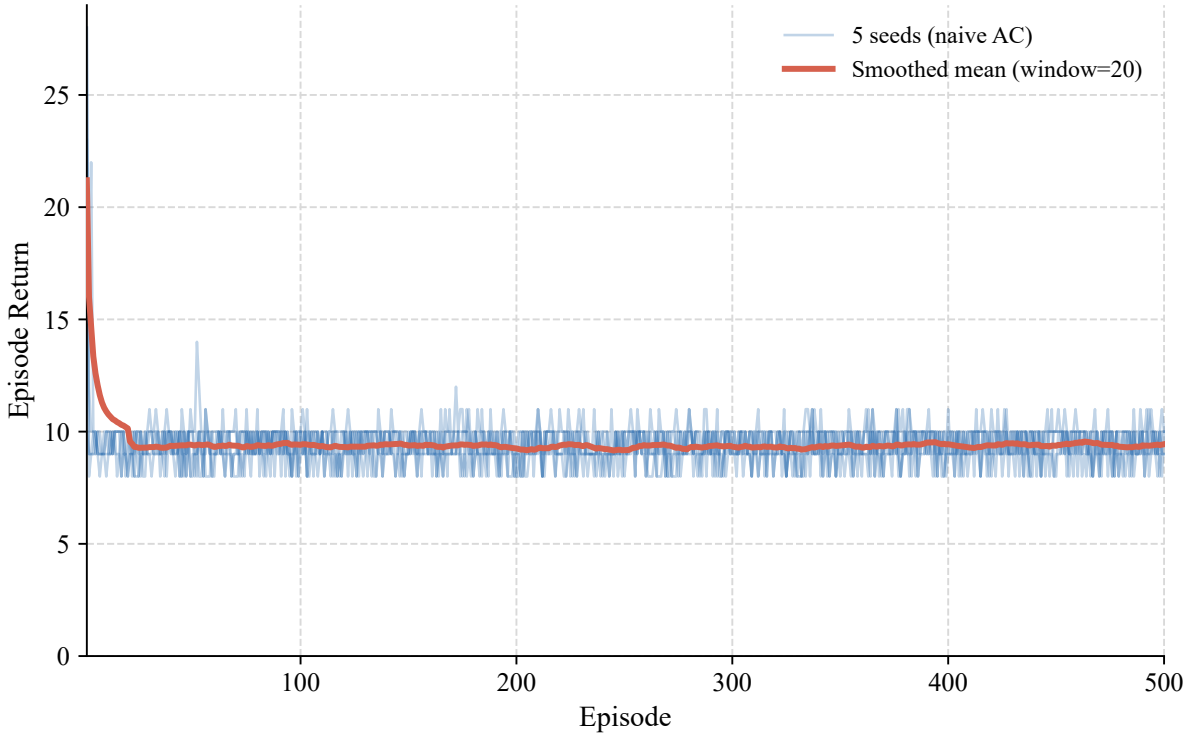


Figure 1: Naive Actor-Critic ($\alpha = 5 \times 10^{-3}$, no entropy, no gradient clipping) on CartPole-v1, 5 random seeds: the return is pinned at ≈ 9 throughout; the maximum return of CartPole-v1 is 500.

Experimental evidence

Figure 1 shows naive Actor-Critic on CartPole-v1 across 5 seeds: the return pins at ≈ 9 (CartPole-v1’s maximum is 500) from about episode 20, with zero progress over 500 episodes. The five curves nearly overlap — not randomness but the same bad fixed point in every seed.

Figure 2 gives direct evidence of the collapse: the mean $|\text{advantage}|$ of naive AC plummets from ≈ 3.5 to ≈ 0.1 within the first 50 episodes and hugs zero thereafter — the actor’s gradient signal has vanished. In contrast, A2C’s $|\text{advantage}|$ traces a **bell curve** — rising as the policy improves (the critic cannot keep up), then smoothly falling as convergence nears — the healthy signature of an online learning system.

Generalized Advantage Estimation (GAE)

The mathematical frame of the bias–variance trade-off

The one-step TD error δ_t is high-bias (when V_ϕ is inaccurate); the Monte Carlo return $G_t - V(s_t)$ is high-variance (longer trajectories, more randomness). Can we find a tunable middle point?

Schulman et al. (2015) proposed the **Generalized Advantage Estimator (GAE)**:

$$\hat{A}_t^{\text{GAE}(\gamma, \lambda)} = \sum_{l=0}^{\infty} (\gamma \lambda)^l \delta_{t+l}$$

Each $\delta_{t+l} = r_{t+l} + \gamma V(s_{t+l+1}) - V(s_{t+l})$ is the TD error at step $t+l$, with $\lambda \in [0, 1]$ the interpolation parameter.

Derivation: GAE as a geometric series of TD errors

Expanding the first n terms:

Fig 3. Advantage Collapse: $|\text{advantage}| \rightarrow 0$ Freezes the Actor

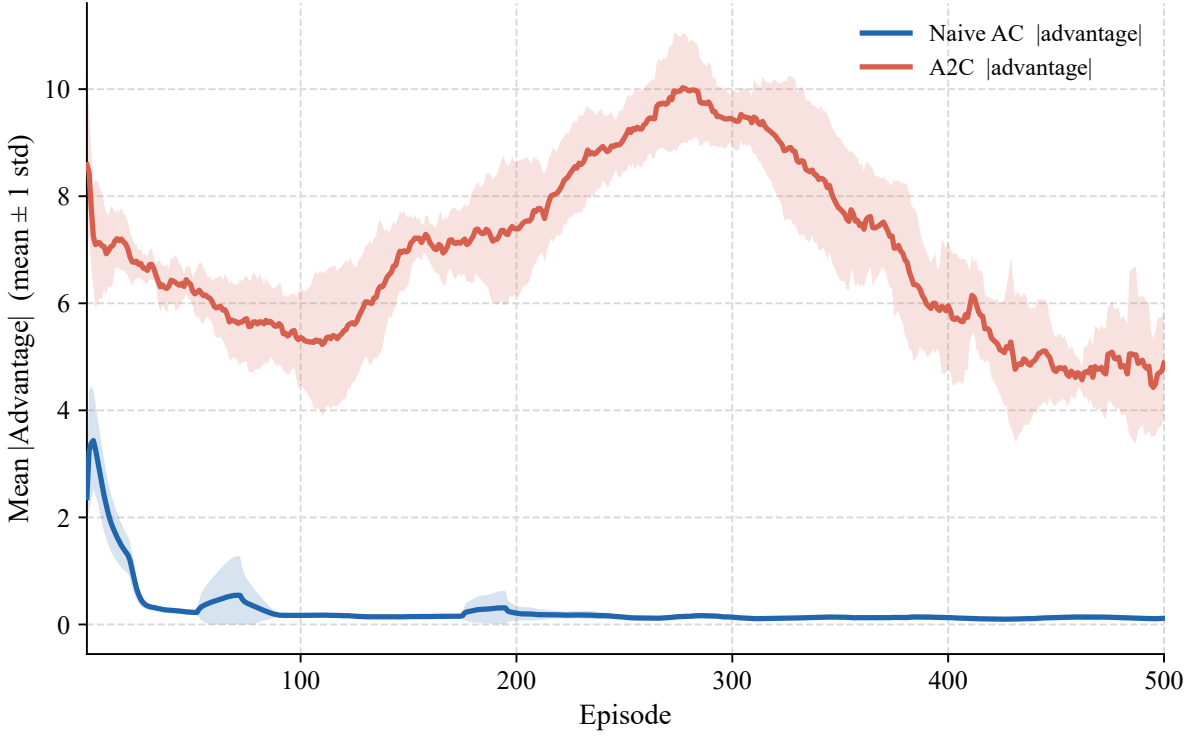


Figure 2: Mean per-episode $|\hat{A}_t|$ (5 seeds, mean $\pm$ 1 std). Blue (naive AC) hits zero within 50 episodes; red (A2C) maintains nonzero signal with a bell shape reflecting the persistent catch-up dynamics between critic and actor.

$$\hat{A}_t^{(n)} = \sum_{l=0}^{n-1} (\gamma\lambda)^l \delta_{t+l} = \delta_t + \gamma\lambda \delta_{t+1} + (\gamma\lambda)^2 \delta_{t+2} + \dots$$

The two extremes:

- $\lambda = 0$: $\hat{A}_t = \delta_t$ (one-step TD), **high bias, low variance**, entirely dependent on V_ϕ 's accuracy;
- $\lambda = 1$: $(\gamma\lambda)^l = \gamma^l$, summing all future TD errors; telescoping gives

$$\hat{A}_t^{\lambda=1} = \sum_{l=0}^{T-t-1} \gamma^l \delta_{t+l} = \underbrace{\sum_{l=0}^{T-t-1} \gamma^l r_{t+l}}_{G_t} - V(s_t) + \gamma^{T-t} V(s_T) \approx G_t - V(s_t), \quad (\text{terminal } V(s_T) \approx 0)$$

i.e. REINFORCE's baseline advantage: **low bias, high variance**.

λ	extreme case	bias / variance
0	$\hat{A}_t = \delta_t$ (one-step TD)	high bias, low variance
1	$\hat{A}_t = G_t - V(s_t)$ (Monte Carlo)	low bias, high variance
$\lambda \in (0, 1)$	geometric interpolation	tunable trade-off

Why geometric decay is sensible: near-term TD errors have a higher signal-to-noise ratio (V_ϕ predicts near states better); distant ones accumulate more randomness. Weighting the near term more via $(\gamma\lambda)^l$ is an implicit exponential smoothing of the gradient estimate.

Using GAE in practice

GAE needs the full sequence of TD errors, so finish a whole rollout first, then recurse backwards from the end:

$$\hat{A}_{T-1} = \delta_{T-1}, \quad \hat{A}_t = \delta_t + \gamma \lambda \hat{A}_{t+1} \quad (t = T-2, \dots, 0)$$

The corresponding **critic target** (the return estimate):

$$\hat{R}_t = \hat{A}_t + V_\phi(s_t)$$

Choosing λ : $\lambda = 0.95$ is the empirical default, working well on most continuous and discrete control tasks (Schulman et al. 2015, 2017).

Advantage Actor-Critic (A2C): four fixes

Patching the four faults one by one

1. **Fix 1 — full-episode rollout + GAE**

Collect a complete trajectory before computing all advantages, replacing one-step δ_t by GAE ($\lambda = 0.95$): lower variance than one-step TD, lower bias than Monte Carlo.

2. **Fix 2 — two-timescale learning rates**

Critic learning rate $\alpha_w = 10^{-3}$, actor $\alpha_\theta = 3 \times 10^{-4}$ ($\alpha_w/\alpha_\theta \approx 3$). The critic learns faster each round, so the actor’s next update uses a more accurate V_ϕ .

3. **Fix 3 — entropy regularization**

Add an entropy term to the actor loss to prevent premature collapse to a deterministic distribution:

$$L_{\text{actor}} = -\hat{A}_t \log \pi_\theta(a_t | s_t) - \beta \mathcal{H}[\pi_\theta(\cdot | s_t)], \quad \beta = 0.01$$

4. **Fix 4 — gradient clipping**

Clip gradients by max-norm before stepping (clip = 0.5), preventing one oversized update from blowing up the parameters.

The full algorithm

1. **Rollout (no gradients):** run one episode with π_θ and V_ϕ ; record $(s_t, a_t, r_t, V_\phi(s_t))_{t=0}^T$;

2. **Compute GAE advantages:** recurse backwards from the end; compute return targets $\hat{R}_t = \hat{A}_t + V_\phi(s_t)$;

3. **Normalize advantages:** $\hat{A}_t \leftarrow (\hat{A}_t - \bar{A})/(\sigma_A + \varepsilon)$, stabilizing gradient magnitudes;

4. **Critic update (higher learning rate):**

$$L_{\text{critic}} = \frac{1}{2} \left\| V_\phi(s_t) - \hat{R}_t \right\|^2, \quad \text{step with } \alpha_w \text{ after gradient clipping;}$$

5. **Actor update (lower learning rate):**

$$L_{\text{actor}} = -\hat{A}_t \log \pi_\theta(a_t | s_t) - \beta \mathcal{H}[\pi_\theta(\cdot | s_t)], \quad \text{step with } \alpha_\theta \text{ after gradient clipping.}$$

Results

Figure 3 shows 5-seed mean ± 1 std curves: A2C rises steadily from about episode 50, reaching a mean of ≈ 310 at episode 500 while still climbing (not fully converged); naive AC hugs the x axis; the two separate completely after episode 50.

Revisit Figure 2’s A2C curve (the red bell): the advantage keeps growing for the first ~ 280 episodes — the phase of fast return growth in Figure 3, where the policy improves faster than the critic can track; afterwards it falls smoothly as the critic catches up and the return curve’s slope eases. **The critic–actor catch-up dynamic is exactly the healthy state the two-timescale design wants to maintain.**

Fig 2. A2C vs. Naive Actor-Critic: Convergence Comparison

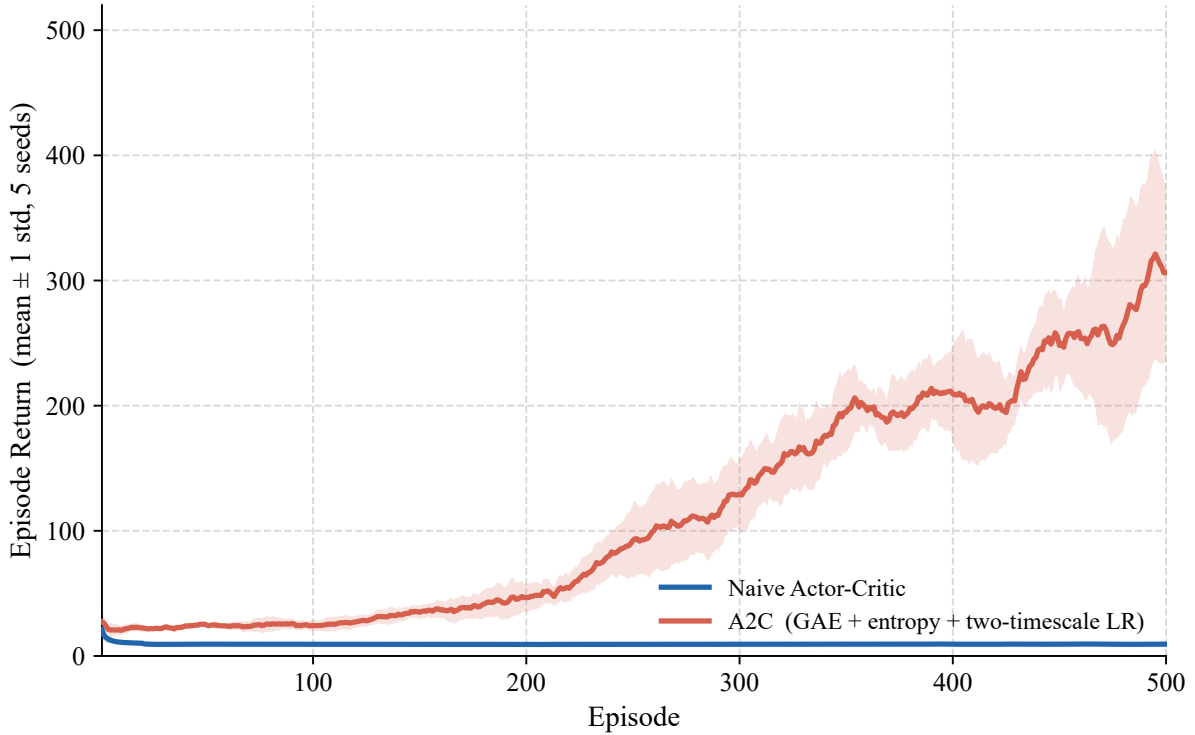


Figure 3: A2C vs. naive Actor-Critic on CartPole-v1 (5 seeds, mean $\pm$ 1 std, identical network architecture; only training differs).

Algorithm comparison

Method	Advantage estimate	Update timing	Bias / variance
REINFORCE	G_t	end of episode	low bias, very high variance
REINFORCE + Baseline	$G_t - V(s_t)$	end of episode	low bias, high variance (improved)
naive AC	δ_t	online, per step	high bias, low variance (but collapses)
A2C	$\hat{A}^{\text{GAE}(\gamma, \lambda)}$	batched after episode	tunable via λ , stable

Why not just use Monte Carlo returns in A2C?

Using pure MC returns G_t as the critic target reduces A2C to REINFORCE with baseline — bootstrap’s variance advantage disappears. On **short-episode** tasks like CartPole-v1 MC is usable, but on continuous-control (high-dimensional action spaces such as joint angles and velocities) plus long episodes, G_t ’s variance becomes unmanageable. GAE’s λ interpolates continuously over the bias–variance trade-off instead of choosing one extreme.

Why should the critic’s learning rate exceed the actor’s?

The two-timescale theorem (Konda & Tsitsiklis 2003): Actor-Critic convergence requires

$$\sum_k \alpha_\theta(k) = \infty, \quad \sum_k \alpha_w(k) = \infty, \quad \frac{\alpha_\theta(k)}{\alpha_w(k)} \rightarrow 0 \quad (k \rightarrow \infty)$$

Intuition: if actor and critic move at the same speed, the critic forever chases a moving target (the policy keeps changing). A faster critic approximately converges to the current policy’s V^π before the actor’s next update, guaranteeing the actor a “trustworthy evaluation of the current policy” — gradient directions it can trust.

Where this chapter sits in the RL landscape

Actor-Critic is the bridge from basic algorithms to modern deep RL:

1. **The TD route (value-based):** MAB $\rightarrow$ MDP / Bellman equations $\rightarrow$ TD(0) / Q-Learning $\rightarrow$ DQN. DQN learns $Q(s, a)$ — a pure critic with no explicit actor; it cannot handle continuous action spaces.
2. **The MC route (policy-based):** REINFORCE optimizes π_θ directly — a pure actor with no critic; enormous variance, low sample efficiency.
3. **The Actor-Critic route (fusion):** this chapter merges the two — the actor optimizes the policy while the critic supplies a low-variance baseline. GAE tunes the bias-variance trade-off $\rightarrow$ **PPO** (next chapter) adds a clip constraint on the AC framework to bound each policy update $\rightarrow$ **SAC** pushes entropy regularization to a firmer theoretical footing with automatic temperature tuning.
4. **The shadow of offline RL:** in offline methods such as IQL (Implicit Q-Learning) the critic still needs accurate value estimates; the two-timescale principle reappears in another guise — **how to keep the critic from overfitting out-of-distribution states when coverage is thin** is precisely the question offline RL answers.

Remove PPO's clip objective, keeping GAE + entropy + gradient clipping — how different is what remains from this chapter's A2C? What exactly does PPO's clip prevent?

References

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